

# Counting Sources, Not Bursts: Source Clustering in the Repeater/Non-Repeater Dispersion-Measure Comparison of CHIME/FRB Catalog 1

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## Abstract

We report a preregistered, independent re-analysis of one comparison in the first CHIME/FRB fast radio burst catalog: do repeating and apparently non-repeating bursts differ in dispersion measure (DM), intrinsic pulse width and spectral bandwidth? On 497 analysed bursts (59 bursts from 18 repeating sources, 438 non-repeaters), width and bandwidth differ clearly (Kolmogorov–Smirnov  $p < 10^{-8}$ ), as the catalog paper reports. The DM comparison, however, is highly significant at burst level ( $p = 2.0 \times 10^{-10}$ ) where the catalog paper, which uses one burst per repeating source, reports consistent distributions. A post-hoc diagnostic explains the difference: bursts from the same repeating source share essentially one DM excess (intraclass correlation 0.99999), so the 59 repeater bursts behave like about 7 independent observations. At the independent unit, the 18 sources, the DM difference is marginal (KS  $p = 0.041$ ; Mann–Whitney  $p = 0.026$ ) and not significant at  $p < 0.01$ , while the width and bandwidth differences survive every source-level test. We also correct an error in this project’s earlier documentation, which wrongly attributed the discrepancy to two prolific sources. The result is a statement about the unit of analysis, not about whether repeaters form a distinct population.

## 1 Question

The first CHIME/FRB catalog [1] lists 536 bursts detected between 2018-07-25 and 2019-07-01, 62 of them from 18 previously known repeating sources. Its repeater/non-repeater comparisons use each repeating source’s first-detected burst, so that every source counts once. We asked a narrow question with the same public data: does that comparison hold if every analysed burst is used instead, under a protocol fixed before any result was generated? The question is one of statistical design, since repeat bursts from one source are not independent draws from the population of sources [3]. We make no claim that repeaters and non-repeaters are, or are not, physically distinct populations.

## 2 Data and preregistration

We used the machine-readable catalog table at CDS VizieR ([JApJS/257/59/table2](https://vizier.cds.unistra.fr/viz-bin/VizieR?source=JApJS+257+59+table2)), selecting one row per burst. Structural checks run at fetch time confirm 536 bursts and 62 repeater bursts from

18 sources. Following the catalog’s own flag, 39 bursts from non-nominal telescope operation were excluded, leaving 497 bursts: 59 from 18 repeating sources and 438 non-repeaters. The protocol (committed before the registry) fixed: the DM excess after Galactic subtraction with NE2001 as the primary measure (raw fitburst DM and YMW16 as robustness checks), intrinsic pulse width, and bandwidth = high – low frequency; a two-sample Kolmogorov–Smirnov (KS) test as the primary test with Anderson–Darling [4] and a 10,000-resample bootstrap of the median difference as checks. Two hypotheses were stated: **H1**, DM distributions are indistinguishable; **H2**, width and bandwidth differ. The registry regenerates identically from a fresh download.

### 3 Preregistered results

| Measure                        | $n$ (rep./non) | KS $D$ | KS $p$                | Median rep. | Median non-rep. |
|--------------------------------|----------------|--------|-----------------------|-------------|-----------------|
| DM excess, NE2001 (primary)    | 59/438         | 0.457  | $2.0 \times 10^{-10}$ | 157.0       | 510.7           |
| DM, raw fitburst               | 59/438         | 0.479  | $2.0 \times 10^{-11}$ | 349.7       | 572.8           |
| DM excess, YMW16               | 59/438         | 0.487  | $8.1 \times 10^{-12}$ | 164.6       | 498.7           |
| Pulse width (ms)               | 59/438         | 0.433  | $2.4 \times 10^{-9}$  | 2.00        | 0.93            |
| Width, excl. limit-flagged     | 59/415         | 0.425  | $5.9 \times 10^{-9}$  | 2.00        | 0.97            |
| Bandwidth (MHz)                | 59/438         | 0.522  | $1.3 \times 10^{-13}$ | 206.0       | 346.1           |
| Bandwidth, excl. limit-flagged | 59/415         | 0.529  | $6.4 \times 10^{-14}$ | 206.0       | 351.5           |

Table 1: Preregistered burst-level comparisons (DM in  $\text{pc cm}^{-3}$ ). Anderson–Darling  $p \leq 0.001$  in every row.

**H2 is confirmed:** repeaters have wider pulses and narrower bandwidth, robust to every variant. **H1 is falsified** at burst level, under all three DM conventions, with repeaters at lower DM (median DM excess 157.0 against 510.7  $\text{pc cm}^{-3}$ ; bootstrap 95% interval of the difference 167.4–389.4).

### 4 Post-hoc diagnostics: source clustering

The H1 result differs from the catalog paper’s, and the catalog paper counts sources, not bursts. We therefore ran, after the primary result, four fixed diagnostics (not preregistered; all reported, for all three measures): (i) the intraclass correlation (ICC, one-way random effects [5]) of each measure within sources, with the Kish design effect and effective sample size [6]; (ii) a source-level test in which each repeating source is collapsed to the median of its bursts; (iii) 10,000 random draws of one burst per source (the catalog paper’s first-detection rule is one such draw); (iv) the burst-level KS test with each source removed in turn.

|                                       | DM excess (NE2001)                         | Pulse width                                | Bandwidth                                     |
|---------------------------------------|--------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Within-source ICC                     | 0.99999                                    | 0.19                                       | 0.28                                          |
| Design effect (Kish)                  | 8.36                                       | 2.39                                       | 3.07                                          |
| Effective $n$ of 59 bursts            | 7.1                                        | 24.7                                       | 19.2                                          |
| Source-level KS $p$ (18 sources)      | 0.041                                      | $1.3 \times 10^{-5}$                       | $4.4 \times 10^{-6}$                          |
| Source-level Mann–Whitney $p$         | 0.026                                      | $4.7 \times 10^{-6}$                       | $1.4 \times 10^{-5}$                          |
| 1 burst/source, KS $p$ : median (max) | 0.041 (0.041)                              | $2.6 \times 10^{-4}$ (0.046)               | $1.5 \times 10^{-5}$ ( $2.5 \times 10^{-3}$ ) |
| Leave-one-source-out KS $p$           | $2 \times 10^{-11}$ – $4.4 \times 10^{-5}$ | $3 \times 10^{-10}$ – $1.6 \times 10^{-7}$ | $2 \times 10^{-14}$ – $8 \times 10^{-11}$     |

Table 2: Post-hoc source-clustering diagnostics (seed 20261009). Cluster sizes: 19, 8, 3, 3, twelve sources with 2, two with 1.

**Findings.** *DM:* within a repeating source the DM excess varies by only a few  $\text{pc cm}^{-3}$  against hundreds between sources, so the ICC is 0.99999 and the 59 repeater bursts carry the information of about 7 independent observations. Every one-burst-per-source draw gives the same  $p = 0.041$ , since each source has effectively one DM. At the independent unit, the difference is marginal: KS  $p = 0.041$  and Mann–Whitney  $p = 0.026$ , nominally significant at  $\alpha = 0.05$  but not at the catalog paper’s  $p < 0.01$  convention. The median of the 18 source medians is 358.7 against 510.7  $\text{pc cm}^{-3}$  for non-repeaters. The burst-level significance therefore comes from counting each source’s DM several times, with weights set by burst count, not from a few dominant sources: removing any single source, including the most prolific (19 bursts), leaves the burst-level difference significant ( $p \leq 4.4 \times 10^{-5}$ ). *Width and bandwidth:* ICCs are low to moderate, and every source-level test, every random one-burst-per-source draw ( $p \leq 0.046$  for width,  $\leq 2.5 \times 10^{-3}$  for bandwidth) and every leave-one-out test remains significant. The replicated contrast is not an artifact of clustering.

## 5 Discussion

The burst-level DM discrepancy is real in the data but is an artifact of the unit of analysis: the catalog paper’s choice to count sources is the statistically appropriate one for DM, and under it the repeater–non-repeater difference is at most marginal in this catalog. This does not show that repeaters and non-repeaters share a DM distribution: with 18 sources the test has little power, the source-level  $p$ -values sit near conventional thresholds, and a different DM convention, exclusion or added source could move them across. A later CHIME/FRB study of 25 newly discovered repeaters reports lower mean DM for repeaters [2]; that is directionally consistent with our median difference but uses a different sample and does not validate any burst-level  $p$ -value here. The accurate summary is that the DM evidence is sample- and selection-dependent. A practical lesson follows for catalog-level comparisons: report the number of independent units, and report the ICC when units contribute several observations.

## 6 Correction to earlier project documentation

This project’s earlier report and web pages stated that two repeaters (FRB 20180916B and FRB 20180814A) supplied 33 and 20 of the 59 repeater bursts, “90%” of the sample, and called that the likely driver of the discrepancy. While preparing this paper we recounted from the verified catalog: the counts are 19 and 8, i.e. 27 of 59 (46%). The statement was wrong and has been corrected (erratum in the repository report). No registry value, test statistic or hypothesis disposition changed.

## 7 Limitations

One catalog, one telescope and band (400–800 MHz), one year. Repeater is a detection-window label, not a guaranteed physical category. CHIME/FRB selection effects in DM, width and sky position are not modelled; catalog-derived columns are used as given. Multi-component bursts are reduced to their first sub-burst. The clustering diagnostics are post-hoc, descriptive, and use small numbers of sources; we claim an explanation of the discrepancy, not a measurement of any population difference.

## Reproducibility and AI assistance

Code, protocol, frozen registries, the clustering diagnostics and tests:

<https://github.com/lindgreendavid/frb-atlas>

Interactive laboratory:

<https://frb-atlas-interactive.lindgreendavid.workers.dev>

The analysis code, tests and drafts of this text were produced with substantial assistance from an AI system (Claude, Anthropic); the author is responsible for the content. The protocol commit precedes the registry, and the post-hoc diagnostics are in a separate, labelled file.

## References

- [1] CHIME/FRB Collaboration (M. Amiri et al.), The first CHIME/FRB fast radio burst catalog, *Astrophys. J. Suppl.* **257**, 59 (2021), doi:10.3847/1538-4365/ac33ab; erratum **264**, 53 (2023) (all-sky rate only), doi:10.3847/1538-4365/acb54c.
- [2] CHIME/FRB Collaboration, CHIME/FRB discovery of 25 repeating fast radio burst sources, *Astrophys. J.* (2023), doi:10.3847/1538-4357/acc6c1.
- [3] S. H. Hurlbert, Pseudoreplication and the design of ecological field experiments, *Ecol. Monogr.* **54**, 187–211 (1984).
- [4] F. W. Scholz and M. A. Stephens,  $K$ -sample Anderson–Darling tests, *J. Am. Stat. Assoc.* **82**, 918–924 (1987).
- [5] P. E. Shrout and J. L. Fleiss, Intraclass correlations: uses in assessing rater reliability, *Psychol. Bull.* **86**, 420–428 (1979).
- [6] L. Kish, *Survey Sampling*, Wiley (1965).